

Calculation of Steam Boiler OEE

Availability, Performance and Quality Factors



Boiler OEE

- Total good hours=738hrs
- Mechanical breakdown=42.92hrs,
- Electrical breakdown=12.24hrs
- Electronics/safety device breakdown=3.01hrs
- Total breakdown= Mechanical + Electrical + Electronics/Safety Breakdown
- Total breakdown = 58.18hrs
- Setup & Adjustments = 8.5hrs
- Total loss = Total Breakdown + Setup & Adjustments
- = 58.18 + 8.5 = 66.68hrs

Boiler OEE

- Boiler Availability
 - Total good hours=738hrs
 - Net Loss = Total Good hours – Total Loss
 - $= 738 - 66.68$
 - $= 671.32$
 - Availability = Net Loss/Total Good Hours = $671.32/738$
 - $= 0.9096$
 - $= 90.96\%$

Boiler OEE

- Boiler Performance
 - Net Loss = 671.32hrs
 - If Minor Stoppages = 50hrs
 - Performance = (Net Loss – Minor Stoppages)/Net Loss
 - = $(671.32 - 50)/671.32$
 - = $621.32/671.32$
 - = 0.9251
 - = 92.51%

Boiler OEE

- Boiler Quality

- Quality = Good Steam Produces/Total Steam Produced
- Quality = (Total steam produced-Defected steam)/Total steam produced
- Total Steam Produced = 7380 tons
- Defected Steam = Total Breakdowns * Steam Produced Per hour
- Defected Steam = 581.8 tons
- Good Steam Produced = Total Steam Produced – Defected Steam Produced
- $= 7380 - 581.8$
- $= 6798.2$ tons
- Quality = $6798.2/7380 = 0.9211 = 92.11\%$

Boiler OEE

- $OEE = Availability * Performance * Quality$
- $= 0.9096 * 0.9251 * 0.9211$
- $= 0.7745$
- $= 77.49\%$